

Information Security

BCA — Semester 5 — — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

- 1. Consider $p=11$ and $q=7$ in a RSA cryptosystem. i. What is a public key pair (e, n) ? ii. What is a private key pair (d, n) ? iii. What is ciphertext for $M=6$?**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

- 2. Discuss how encryption and decryption is done in the DES algorithm.**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

- 3. Define subjects, objects and access rights in access control with suitable examples. How role based access control is different from attribute based access control?**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

- 4. How online and offline dictionary attacks are done in password based authentication systems?**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

- 5. Describe the roles of relying parties, attribute providers and identity providers in Open Identity Trust Framework.**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

- 6. Define zombies, rootkits and Trojans.**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

- 7. Briefly describe the status of cyber law in Nepal.**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Discuss various methods of risk treatment during security risk analysis.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What is the use of S-box in DES? Illustrate S-box operation with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. How hash value is generated by the SHA-2 hash function.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Write Rabin Miller Algorithm for primality testing. Test whether 341 is prime or not using the algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Define interception, repudiation and incapacitation with examples.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.